Parametric Equations: Introduction

Questions 1–21

Question 1. Prerequisite knowledge: Those questions that involve trigonometry use radians For each of the following, complete the table, plot the coordinates and sketch the curve for $-2 \leq t \leq 2$.

t	-2	-1	0	1	2
x					
y					

a) $x = t^2$ and $y = 2t$
b) $x = t + 2$ and $y = -2t^2 + 1$
c) $x = t(t + 2)$ and $y = t^2(t - 1)$
d) $x = t(1 - t)$ and $y = 2t^2$
e) $x = \frac{4}{2+t^2}$ and $y = t(t + 1)$

Question 2. Given that $x = t$, find y in terms of t for each of the following pairs of parametric equations:
a) $y = 2x^2$
b) $y = x^2 + 4$
c) $y = e^x + 3$
d) $y - 2x = 8$
e) $\frac{x^2}{y} = 1$
f) $x^3y = -10$
g) $x^2 + y^2 = 1$

Question 3. Find the cartesian equation (with y as the subject) for each of the following curves:

a) $x = 2t$ and $y = t^2$
b) $x = t - 1$ and $y = 2t + 3$
c) $x = t - 3$ and $y = e^t - t$
d) $x = -2t$ and $y = t^2 \ln t$

Question 4. Find the cartesian equation (with x as the subject) for each of the following curves:

a) $x = t^2$ and $y = t - 1$
b) $x = t^3 - t$ and $y = \frac{t}{2}$
c) $x = t^5$ and $y = 3t$
d) $x = e^{t-1}$ and $y = t + 5$

Question 5. For each of the following curves, find the coordinates of intersections with the axes:

a) $x = t + 1$ and $y = t - 3$
b) $x = \cos t$ and $y = 2\sin t$ for $0 \leq t < 2\pi$
c) $x = 2\cos t$ and $y = \sin 3t$ for $0 \leq t < 2\pi$
d) $x = t + 5$ and $y = \ln(t + 1)$
e) $x = t^3 - 3t^2 - 6t + 8$ and $y = e^{-2t}$

Question 6. A curve is given by parametric equations

$$x = t^2 \quad \text{and} \quad y = t^3$$

Given the curve may be written in the form $y^2 = x^k$, state the value of k.

Question 7. Find the cartesian equation for each of:

a) $x = t + \frac{1}{t}$ and $y = t - \frac{1}{t}$

b) $x = t - \frac{2}{t}$ and $y = t + \frac{2}{t}$

c) $x = 7t + \frac{4}{t}$ and $y = 7t - \frac{4}{t}$

Question 8. A curve is given by parametric equations

$$x = \frac{1}{t} - 1 \quad \text{and} \quad y = \frac{2+3t}{1+3t}$$

Show that the cartesian equation of the curve can be written in the form $y = \frac{ax+b}{x+4}$, $x \neq -4$ where a and b are integers to be found.

Question 9. Find a pair of parametric equations for each of the following cartesian equations:

a) $4x^2 + 9y^2 = 1$
b) $3x^2 + 2y^2 = 1$
c) $5x^2 + \frac{1}{2}y^2 = 1$
d) $4x^2 + \frac{2}{3}y^2 = 2$

Question 10. Show that each of the following may be written in the form

$$(x - p)^2 + (y - q)^2 = k$$

where p, q and k are integers and $0 \leq \theta < 2\pi$:

a) $x = 3\cos\theta$ and $y = 3\sin\theta$
b) $x = 5\sin\theta$ and $y = -5\cos\theta$
c) $x = \cos\theta + 1$ and $y = \sin\theta - 3$
d) $x = \sin\theta - 4$ and $y = \cos\theta + 8$

Question 11. A fireworks display is planned so that a sequence of firework bursts follow a trajectory in the sky, in metres, such that

$$x = 10\cos t \quad \text{and} \quad y = 15 + 5\sin t$$

for $0 < t \leq 2\pi$ where t represents the position of a burst of a firework at time t seconds. Find

a) the cartesian equation of the trajectory traced by the fireworks.
b) the maximum height reached by the fireworks in one cycle of motion.
c) the value of t which provides the maximum height in part b).
d) the period of the fireworks sequence.

Question 12. A curve is defined for $0 \leq \theta < 2\pi$ by

$$x = 2k + 2k\cos\theta \quad \text{and} \quad y = 8 - 3\sin\theta$$

for $k > 0$.

a) Find a cartesian equation for the curve.

It is given that the curve is a circle. Find

b) the value of k.
c) the centre of the circle.
d) the radius of the circle.

Question 13. Prerequisite knowledge for Q13: Trig Identities Write the following

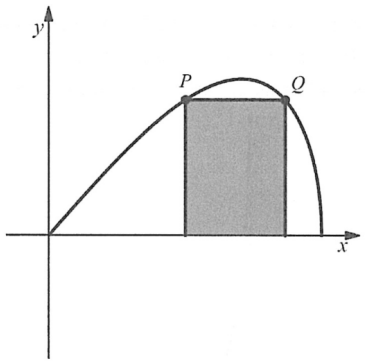
$$x = 2\tan\theta \text{ and } y = 2 + \sec\theta, \quad 0 \leq \theta < 2\pi$$

as a cartesian equation.

Question 14. A curve is defined by the parametric equations

$$x = \sqrt{3}\sin\theta, \quad y = \sin 2\theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.



The points P and Q lie on the curve. The line $x = \frac{3}{2}$ meets the curve at the point Q.

a) Find the coordinates of P and Q.

b) Hence find the exact area of the rectangle bounded by the x-axis and the points P and Q.

Question 15. Find the cartesian form of the curve which satisfies all the points on:

a) $x = \cos\theta$ and $y = \sin^2\theta$ for $0 \leq \theta \leq \pi$

b) $x = \sin\theta$ and $y = \tan\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

Question 16. A curve is defined by

$$x = \frac{t^2+7}{t^2+1} \quad \text{and} \quad y = \frac{6t}{t^2+1}$$

a) Show that the curve defined by $t \in \mathbb{R}$ is a circle with centre $(4, 0)$.

b) Find the radius of the circle.

Question 17. For each of the following parametric curves, find all the points of intersection with the circle given by

$$x^2 + y^2 = 100$$

where $0 \leq \theta \leq 2\pi$. Give your answers in exact form.

a) $x = 10\cos\theta$, $y = 4\sin\theta$

b) $x = 2\cos\theta$, $y = 10\sin\theta$

c) $x = 6\sqrt{2}\cos\theta$, $y = 2\sqrt{46}\sin\theta$

Question 18. Prerequisite knowledge for Q18: Double Angle Formulae A curve is given by parametric equations

$$x = 5 + 2\cos\theta \quad \text{and} \quad y = -1 + 2\cos 2\theta$$

for $0 \leq \theta \leq \pi$.

a) Find the cartesian equation of the curve which satisfies all the points on the curve.

b) State the domain and range of the curve.

The line $y + x = k$ where k is a constant, intersects the curve at two distinct points.

c) State the range of the values of k giving your answer in interval notation.

Question 19. Prerequisite knowledge for Q19–Q21: Domain and Range of Functions Find the largest possible domain and range of:

a) $x = t^2 + 1$ and $y = \sqrt{t}$
b) $x = 2t^2$ and $y = 4 - \sqrt{t}$
c) $x = \sin t$ and $y = \cos t$
d) $x = \sin 3t$ and $y = \cos(t + \frac{\pi}{2})$
e) $x = e^t$ and $y = e^{-2t}$
f) $x = \ln t$ and $y = t^2$

Question 20. By first considering valid values for t, find the largest possible domain and range of the curve defined by $x = \sin\sqrt{t}$ and $y = \frac{1}{t}$.

Question 21. For $0 \leq \theta < 2\pi$ a curve is defined by

$$x = \sin 2\theta \quad \text{and} \quad y = 4\cos(\theta - \tfrac{\pi}{6})$$

a) Find the domain and range of the curve.
b) Find the coordinates where the curve meets the coordinate axes.
c) Hence, sketch the curve.